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TUNIS BUSINESS SCHOOL



Comparative Analysis of Machine Learning and Deep Learning Models for Mushroom Binary Classification

MBA 523: Neural Networks & Deep Learning

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Master's in Business Analytics

1 Introduction

Classifying mushrooms as edible or poisonous is a critical task with significant health implications. This project, *Comparative Analysis of Machine Learning and Deep Learning Models for Mushroom Binary Classification*, explores various approaches to solve this problem using the predefined mushroom features.

We apply and then proceed to evaluate five machine learning models alongside three deep learning models. The study also considers the role of preprocessing techniques in preparing the data for model training and ensuring consistency across algorithms. The models will be further assessed using classification evaluation metrics, allowing for a detailed comparison of the different models' performance. All methodologies are harmonized throughout this project. This analysis aims to provide insights into the suitability and effectiveness of each model, offering a robust comparison for binary classification tasks in a real-world scenario.

2 Exploratory Data Analysis (EDA)

This section will discuss the nature of features, addressing aspects such as missing values, potential biases, and the presence of outliers. It will explore how these factors can impact data analysis, and provide strategies for handling them to ensure more accurate and reliable results.

2.1 Shape of the data

The dataset consists of 54035 rows and 9 columns

2.2 Nature of features

The features used in this dataset are quantitative either int or float. The already cleaned dataset contains **9 columns**: Cap Diameter, Cap Shape, Gill Attachment, Gill Color, Stem Height, Stem Width, Stem Color, Season, and **Target Class** - Is it edible or not? defined with 0 and 1 where 0 refers to edible and 1 refers to poisonous.

2.3 Summary statistics

A quick overview of the dataset is provided in the notebook using summary statistics including counts, means, standard deviations, and quartiles for numerical variables, and the count of

unique categories, most frequent values, and their frequencies for categorical variables. This analysis is part of the *Univariate Analysis*, where each column is examined individually helping us understand the distribution and basic characteristics of the data. This step can also help detect outliers and identify patterns.

2.4 Class Distribution

To better understand the balance between the target classes (edible and poisonous mushrooms), we begin by examining the distribution of the target variable.

The dataset contains **29675** records for *poisonous* and **24360** records for *edible* mushrooms, distributed at **55%** and **45%**, respectively. This indicates that the data is relatively balanced.

Additionally, to gain further insights into the differences between the two classes, we analyzed **the mean feature values grouped by class**. The heatmap below visualizes the average values of key features, providing a clear comparison of the differences between edible and poisonous mushrooms:

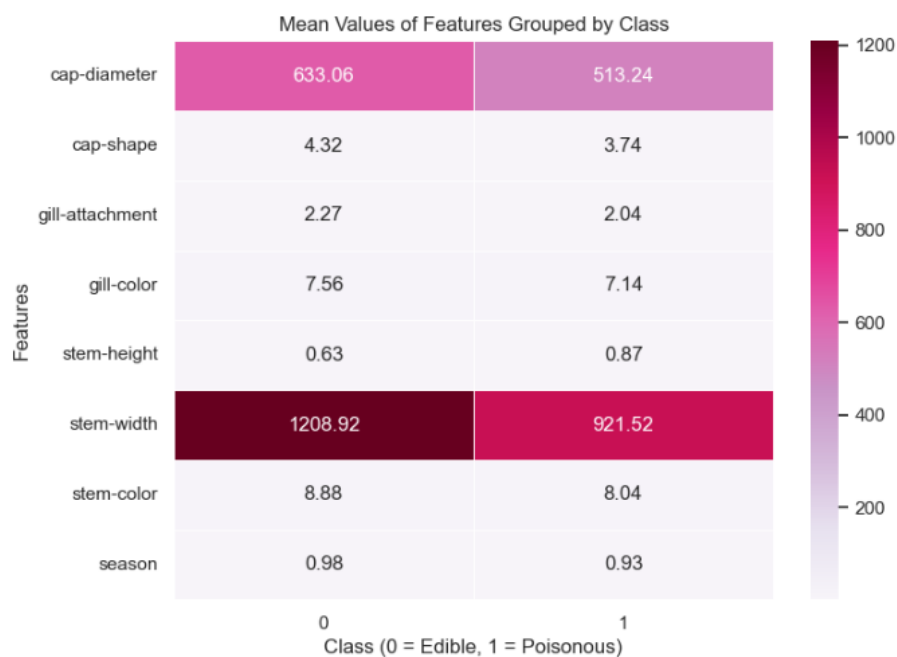


Figure 1: Mean Values of Features Grouped by Class

Derived Insights:

- On average, edible mushrooms have a slightly larger **cap diameter** compared to poi-

sonous mushrooms.

- Poisonous mushrooms have **taller stems** on average.
- Edible mushrooms tend to have **wider stems** on average than poisonous mushrooms.
- Both edible and poisonous mushrooms appear across similar **seasons**, with slight variations, etc...

So, Cap Diameter, Stem Width, and Stem Height show the most significant differences between edible and poisonous mushrooms.

These insights help further distinguish the characteristics of edible and poisonous mushrooms and provide a more nuanced understanding of how these classes differ.

2.5 Feature Correlation

To further explore the relationships between the features also known as *Bivariate Analysis*, a correlation heatmap was generated using *Seaborn*. The heatmap visualizes the correlation matrix, providing a clear understanding of the linear dependencies between different features in the dataset.

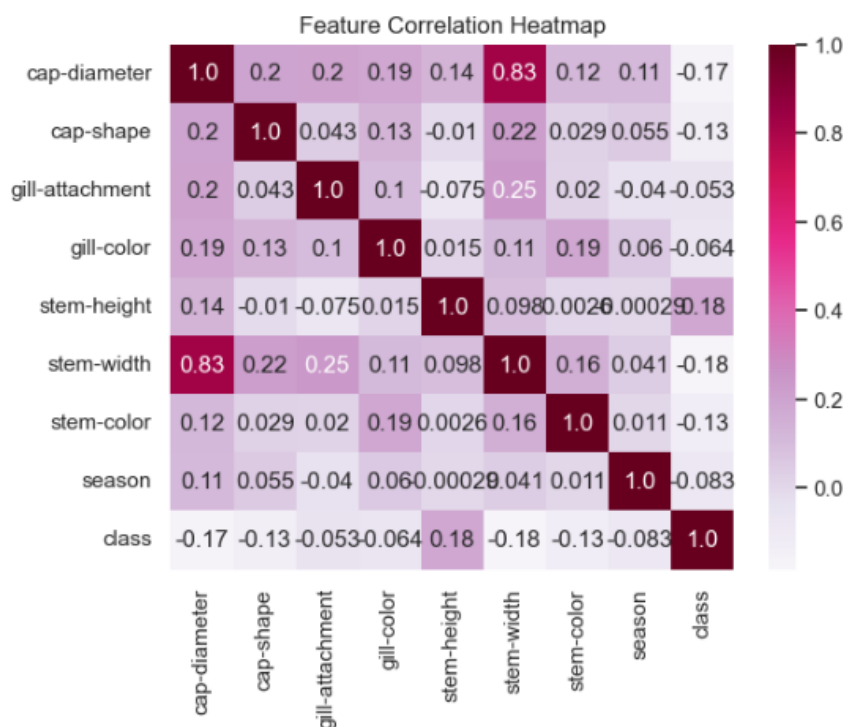


Figure 2: Feature Correlation Heatmap

Derived Insights:

- **cap diameter- stem width:** present a strong positive correlation of **0.83**. Larger cap diameters are associated with thicker stems.
- **cap diameter- cap shape:** present a weak positive correlation of **0.2**. There is a slight relationship between cap diameter and cap shape.
- **cap diameter- class:** present a weak negative correlation of **-0.17**. Larger cap diameters tend to be associated with edible mushrooms.
- **stem width- class:** present a weak negative correlation of **-0.18**. Thicker stems are slightly associated with edible mushrooms.
- **stem height- class:** present a weak positive correlation of **0.18**. Taller stems are slightly associated with edible mushrooms.
- **gill attachment- stem width:** present a weak positive correlation of **0.25**. Certain gill attachments are associated with thicker stems.
- **gill color- stem color:** present a weak positive correlation of **0.19**. Some gill colors are associated with specific stem colors.

2.6 Checking for outliers

- **cap-diameter:** the boxplot shows some extreme values above 1500, meaning there are a few very large caps, Z-scores found 499 outliers, meaning many values are far from the mean and in describe() the Max = 1891 which is way above typical values and even the standard deviation (359.88) is large, meaning cap sizes vary a lot.
Some mushrooms have exceptionally large caps, which might be natural variations and make sense if these mushrooms belong to a specific species.
- **stem height:** doesn't show outliers in the box plot however Z-scores found 887 outliers, meaning many stems are longer than expected. in describe() they showed a max of 3.83 while the mean is 1.08 so it's not too extreme, the high values could just be naturally taller mushrooms.
- **stem width:** The boxplot shows severe outliers, especially near 3500, the Z-scores found 222 outliers, meaning there are extreme values. in describe() the Max = 3569 which is

over double the 75th percentile ($Q3 = 1523$) and the standard deviation (782) is very high, meaning stem widths vary a lot.

The extreme values could indicate measurement errors or unit mismatches (mm vs. cm) which will be corrected after scaling hopefully.

These extreme values indicate either *measurement inconsistencies* or *natural biological variation*. We do not remove them at this stage but acknowledge their presence for further analysis.

3 Data Preprocessing

3.1 Checking for missing data

While the Mushroom dataset is usually complete, it's good practice to check if there are any missing values. After a check we found that there are no missing values so we are good to go.

3.2 Splitting the dataset

We opted for an 80:20 split for the dataset (80% training and 20% testing) which is a standard and effective choice to obtain:

Training set shape: (43228, 8) and *Test set shape:* (10807, 8)

3.3 Feature Scaling

For this project, we opted to use the *MinMaxScaler* for feature scaling as it works well across both Deep Learning and Machine Learning models. Deep learning models, like those built with Keras, benefit from normalized input, and MinMaxScaler helps optimize their performance by scaling features to a fixed range, typically [0, 1].

While KNN, Naive Bayes, and Logistic Regression may perform better with standardization, MinMaxScaler still adapts well to these models.

This approach ensures consistency across all models in the project, *harmonizing the preprocessing steps* without significant drawbacks.

4 Choice of Algorithms

For this Binary Classification task, where the goal is to classify mushrooms as either edible or poisonous, a combination of **five** Machine Learning and **three** Deep Learning models is

selected. This diverse set of algorithms ensures a comprehensive evaluation of model performance and the identification of the most effective approach for this dataset.

Model	Type	Applicability to Mushroom Classification	Strengths	Limitations
K-Nearest Neighbors (KNN)	Instance-Based Learning	Classify based on the proximity of mushrooms to their nearest neighbors using feature similarity	Effective for capturing local patterns; simple to implement	Sensitive to feature scaling and noise; computationally expensive for large datasets
Gaussian Naive Bayes	Probabilistic Model	Assumes feature independence and applies Gaussian distribution for continuous features (e.g., cap diameter)	Fast and interpretable; works well for smaller datasets or when feature independence holds	Struggles with correlated features; less effective for complex feature interactions
Logistic Regression	Linear Model	Evaluates the relationship between features (e.g., gill color, stem width) and mushroom edibility	Interpretable; strong baseline for binary classification problems	Limited in handling non-linear relationships without feature engineering
Decision Tree	Non-Linear Model	Builds interpretable, hierarchical decision rules based on feature thresholds	Handles non-linear patterns; no need for feature scaling; intuitive for feature importance	Prone to overfitting unless pruned; sensitive to noisy data

Random Forest	Ensemble Learning	Uses multiple decision trees to improve classification reliability and reduce overfitting	Robust to noise; captures complex patterns; handles non-linear relationships well	Computationally intensive; less interpretable than single trees
Artificial Neural Network (ANN)	Deep Learning	Learns complex, non-linear relationships between mushroom features and their edibility	Highly flexible and powerful; performs well with large datasets and non-linear interactions	Requires careful hyperparameter tuning; computationally expensive
Convolutional Neural Network (CNN)	Deep Learning (Hierarchical)	Can capture hierarchical patterns	Effective at modeling subtle and complex feature interactions	Less efficient for structured tabular data; may not outperform simpler models for this problem
Recurrent Neural Network (RNN)	Sequential Deep Learning	Not naturally suited for tabular data but could model dependencies among sequential or grouped features	Can explore feature dependencies dynamically; unique approach for structured data	Inefficient for non-sequential data; requires significant adaptation for this dataset

5 Evaluation metrics and Comparative Study

5.1 Choice of Evaluation metrics

For this project, five evaluation metrics are chosen to evaluate the performance of models. The choice of these metrics was derived as they are the optimal evaluation metrics used for classification tasks that provide a quantitative assessment of model performance. They are

presented as follows:

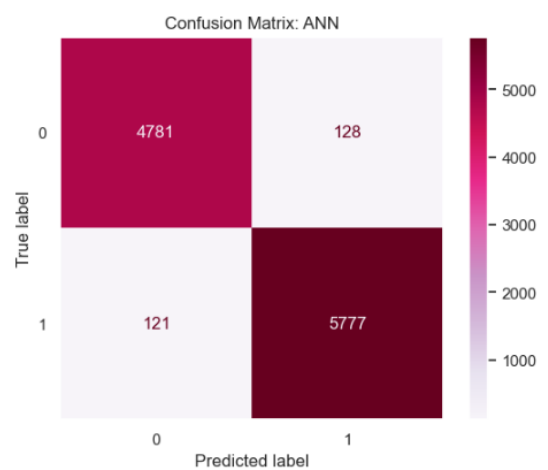
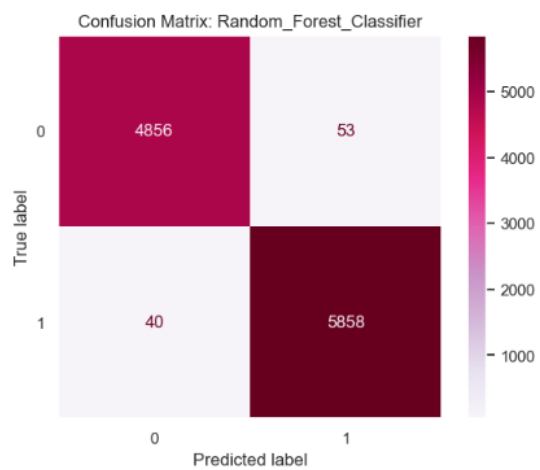
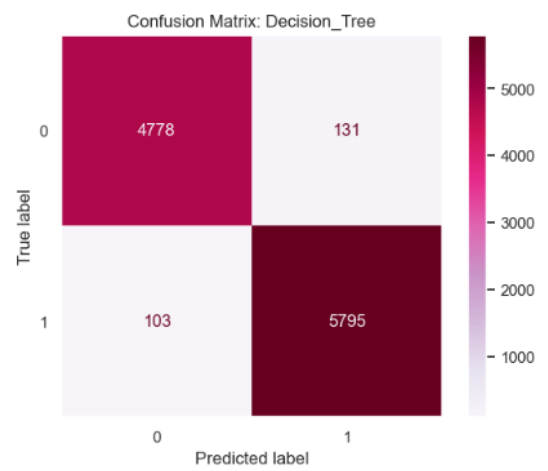
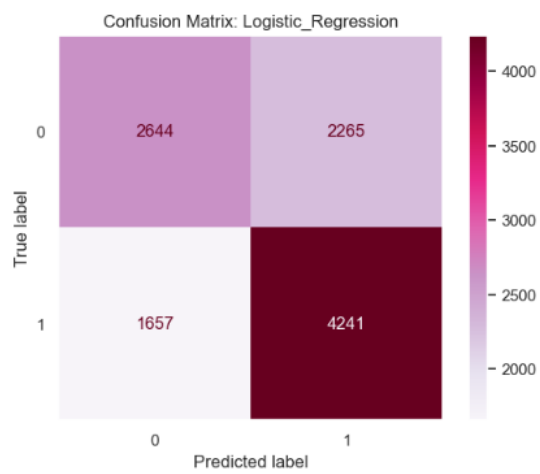
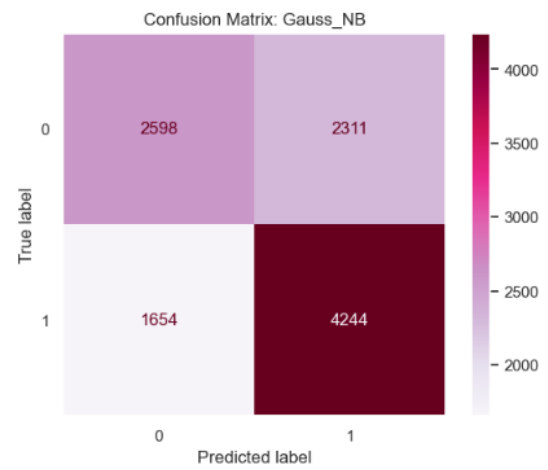
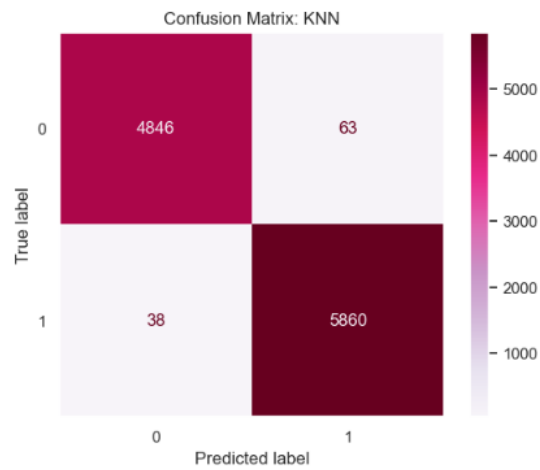
- **Accuracy** which is a commonly used metric for evaluating *Binary Classification* models, reflecting the proportion of correct predictions. It is calculated as:
$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$
- **Precision** measures the proportion of true positive (TP) instances among those predicted as positive by the model. It is calculated as:
$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$
- **Recall** measures the proportion of actual positive instances correctly identified by a model. It is calculated as:
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$
- **F1-score** combines precision and recall, providing a balanced evaluation of a model's performance. It is calculated as:
$$\text{F1-score} = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$$

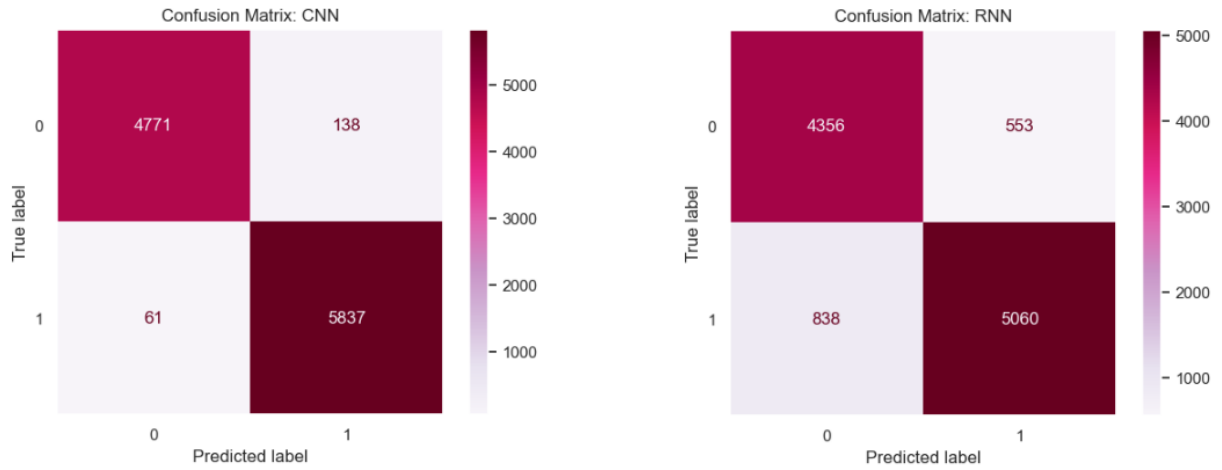
N.B. While the Confusion matrix is not a metric but rather a key tool for evaluating *Binary Classification* algorithms, it shows how well a classification model is performing by comparing its predictions to the actual results and summarizing classification results by showing **correct predictions** for both classes (true positives and true negatives) and **incorrect predictions** (false positives and false negatives) offering insights into the model's performance.

These metrics and tools were developed for each applied model to facilitate a comprehensive comparison of their performance, ensuring consistency and harmony across all models in the project.

5.2 Comparative Study across models

Shown below are the respective *Confusion matrices* of each of the developed models providing an overview of the models' correct and incorrect predictions:





A comparison of each model across the different *predefined metrics* is in the following table:

	Accuracy	Precision	Recall	F1-score
KNN	0.990654	0.989363	0.993557	0.991456
Gaussian Naive Bayes	0.633108	0.647445	0.719566	0.681603
Logistic Regression	0.637087	0.651860	0.719057	0.683812
Decision Tree	0.978347	0.977894	0.982536	0.980210
Random Forest	0.991394	0.991034	0.993218	0.992125
ANN	0.976959	0.978323	0.979485	0.978904
CNN	0.981586	0.976904	0.989658	0.983239
RNN	0.871287	0.901479	0.857918	0.879159

Table 2: Comparative table

Derived Insights:

- CNN performed very well, especially in terms of recall and F1-score. While CNNs are typically used for image or spatial data, they can also work well on sequence-like or tabular data after we consider appropriate reshaping. The ability of CNNs to capture

local patterns (e.g., clusters of features) may be why it outperform simpler models like Logistic Regression and Naive Bayes.

- **RNN** has the lowest performance compared to the other deep learning models. While RNNs are useful for sequence data (time series, text), they are not the best choice for our case of mushroom classification unless an inherent sequential structure is assumed in the data. Also, RNNs are prone to issues like vanishing gradients, which can hinder learning, particularly when working with datasets that don't have a sequential nature.

Comparative Analysis and Key Takeaways:

- **Best Overall Performers:**

For *Machine learning models*, KNN, Random Forest, and Decision Tree achieved the highest accuracy and balanced results making them the top choices for this classification task. These models excelled in the mushroom Binary Classification task, showcasing their ability to handle this dataset effectively with minimal tuning.

As for *Deep Learning models*, both ANN and CNN performed well but didn't outperform the traditional machine learning models, with CNN showing the best recall and F1-score due to its ability to capture spatial relationships in the data (after reshaping). However, RNN performed relatively poorly.

- **Weakest Model:** Gaussian Naive Bayes had the lowest performance across all metrics, likely due to its feature independence assumption, which doesn't hold in our dataset. Also, Logistic Regression showed poor results because it assumes linear relationships between features and might not capture complex interactions between them

N.B. If interpretability and simplicity are key factors, Decision Trees and Random Forests are excellent choices.

If higher performance is the most critical and more computational cost can be afforded, KNN and Random Forest would be ideal.

ANN and CNN are strong candidates but may require further tuning to outperform traditional models. RNN performed better than Gaussian Naive Bayes and Logistic regression even though is not suitable for this task and may require a more sequential or time-based dataset.

To conclude, for this classification task, KNN, Random Forest, and Decision Tree are the top choices for Machine Learning. The Deep Learning models, particularly CNN, showed good performance but did not outperform traditional methods. Logistic Regression, Gaussian Naive Bayes, and RNN performed poorly and are not the ideal models for this dataset.

6 Conclusion and Perspectives

In this project, we successfully implemented and evaluated multiple Machine Learning and Deep Learning models for the Binary Classification of mushroom data. The primary goal was to classify mushrooms as edible or poisonous based on the presence of features and other characteristics. Our models demonstrated promising results, with varying levels of accuracy, precision, and recall, highlighting the importance of **model selection** and **evaluation metrics**.

While the models provided useful insights, several opportunities for improvement and future exploration remain. One potential avenue for enhancing the model performance is *Hyperparameter Tuning* especially for the employed machine learning models and also Fine-tuning parameters such as learning rate, depth of decision trees, or number of layers in neural networks may significantly impact the accuracy and efficiency of the models.

Similarly, *Feature Engineering* plays a critical role in boosting the model's predictive power like creating interaction features or leveraging domain knowledge to generate new features that could uncover hidden patterns that were previously unnoticed.

Moreover, *Data Enrichment* can be adopted, where incorporating additional data sources or creating synthetic data could provide more diverse information, leading to improved generalization capabilities.

Additionally, investigating *imbalanced data handling techniques* to avoid class imbalance even though our data was relatively balanced, such as SMOTE or class-weight adjustments, could potentially enhance the model's ability to handle skewed class distributions and lead to more reliable predictions.